

ACTIVITY 2

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Python Basics

Data Types

```
x = 5
```

```
print(type(x))
```

```
x = str("Hello World")
```

```
print(type(x))
```

```
x = int(20)
```

```
print(type(x))
```

```
x = float(20.5)
```

```
print(type(x))
```

```
x = list(("apple", "banana", "cherry"))
```

```
print(type(x))
```

```
x = dict(name="John", age=36)
```

```
print(type(x))
```

```
<class 'int'>
<class 'str'>
<class 'int'>
<class 'float'>
<class 'list'>
<class 'dict'>
```

Type Casting

```
x = str("s1")
```

```
y = str(2)
```

```
z = str(3.0)
```

```
<class 'str'>
<class 'str'>
<class 'str'>
```

Conditional Statements

```
a = 33
```

```
b = 200
```

```
if b > a:
```

```
    print("b is greater than a")
```

else:

```
print("a is greater than b")
```

```
b is greater than a
```

Python match

```
day = 4
```

```
match day:
```

```
case 1:
```

```
    print("Monday")
```

```
case 2:
```

```
    print("Tuesday")
```

```
case 3:
```

```
    print("Wednesday")
```

```
case 4:
```

```
    print("Thursday")
```

```
case 5:
```

```
    print("Friday")
```

```
case 6:
```

```
    print("Saturday")
```

```
case 7:
```

```
    print("Sunday")
```

```
Thursday
```

While Loop

```
i = 1
```

```
while i < 6:
```

```
    print(i)
```

```
    i += 1
```

```
1
2
3
4
5
```

For Loop

```
fruits = ["apple", "banana", "cherry"]
```

```
for x in fruits:
```

```
    print(x)
```

```
apple  
banana  
cherry
```

Functions

```
def fahrenheit_to_celsius(fahrenheit):  
    return (fahrenheit - 32) * 5 / 9
```

```
print(fahrenheit_to_celsius(77))  
print(fahrenheit_to_celsius(95))  
print(fahrenheit_to_celsius(50))
```

```
25.0  
35.0  
10.0
```

Lambda Functions

```
x = lambda a : a + 10  
print(x(5))
```

```
15
```

Recursion

```
def factorial(n):  
    if n == 0 or n == 1:  
        return 1  
    else:  
        return n * factorial(n - 1)
```

```
print(factorial(5))
```

```
120
```